

Installing JupyterLab (Optional)

[!CAUTION]

This section is optional content. Users utilizing our provided virtual machine image should skip this section.

1. Introduction to JupyterLab

JupyterLab is an interactive development environment—the next-generation successor to the classic Jupyter Notebook. It integrates a wider range of features, supports plugin extensions, and can be operated entirely through a web browser. Simple, convenient, and powerful, it is an exceptional code editing tool well worth using.

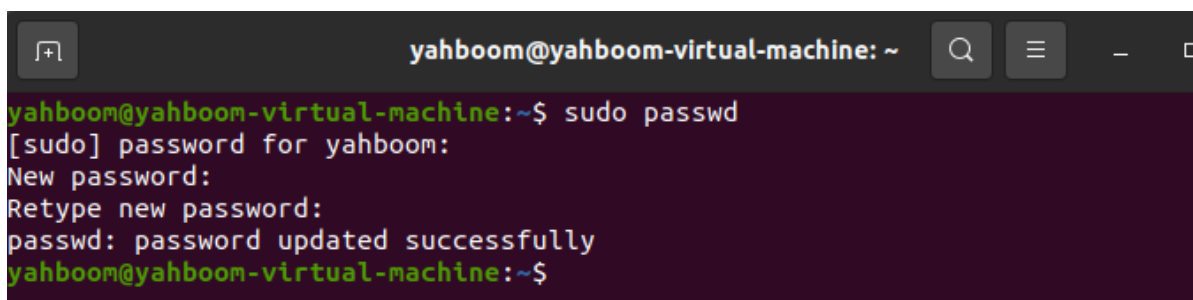
2. Switching to the Root User

Installing JupyterLab requires root privileges. By default, the system's root user account has no password assigned and cannot be directly switched to; therefore, you must set a password for the root user before you can access the account.

Note: You must remember the password you set for the root user here. It is highly recommended to make it identical to your standard user password to ensure you do not forget it easily.

1. Enter the following command to set a password for the root user, then enter the same password twice to confirm.

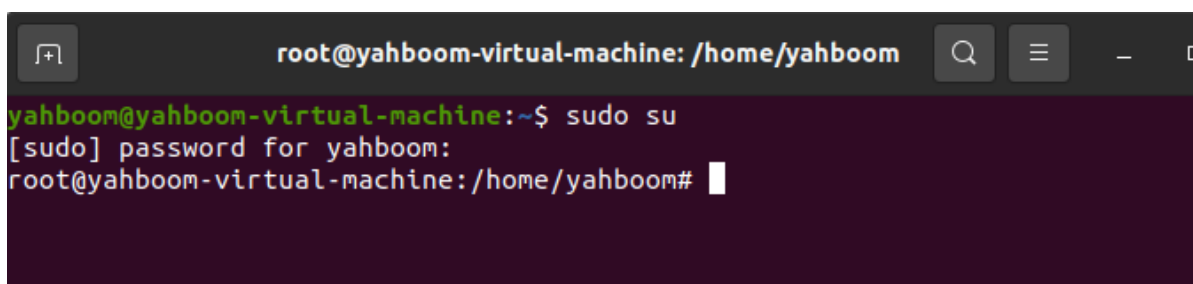
```
sudo passwd
```



```
yahboom@yahboom-virtual-machine: ~  
yahboom@yahboom-virtual-machine:~$ sudo passwd  
[sudo] password for yahboom:  
New password:  
Retype new password:  
passwd: password updated successfully  
yahboom@yahboom-virtual-machine:~$
```

2. Switch to the root user.

```
sudo su
```



```
root@yahboom-virtual-machine: /home/yahboom  
yahboom@yahboom-virtual-machine:~$ sudo su  
[sudo] password for yahboom:  
root@yahboom-virtual-machine: /home/yahboom#
```

As shown in the image above, you have successfully switched to the root user. Additionally, the `$` symbol preceding the command prompt has changed to a `#` symbol.

3. Installing JupyterLab

1. Install the ffi library

```
apt-get install libffi-dev
```

```
root@yahboom-virtual-machine:/home/yahboom# apt-get install libffi-dev
Reading package lists... Done
Building dependency tree
Reading state information... Done
libffi-dev is already the newest version (3.3-4).
libffi-dev set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 277 not upgraded.
root@yahboom-virtual-machine:/home/yahboom#
```

2. Install Jupyter

```
pip3 install -i https://pypi.tuna.tsinghua.edu.cn/simple jupyter
```

```
root@yahboom-virtual-machine:/home/yahboom# pip3 install -i https://pypi.tuna.tsinghua.edu.cn/simple jupyter
Looking in indexes: https://pypi.tuna.tsinghua.edu.cn/simple
Collecting jupyter
  Downloading https://pypi.tuna.tsinghua.edu.cn/packages/83/df/0f5dd132200728a86190397e1ea87cd76244e42d39ec5e88efd25b2abd7e/jupyter-1.0.0-py2.py3-none-any.whl (2.7 kB)
Requirement already satisfied: notebook in /usr/local/lib/python3.8/dist-packages (from jupyter) (6.5.4)
Collecting qtconsole (from jupyter)
  Downloading https://pypi.tuna.tsinghua.edu.cn/packages/24/e2/7f22137bbb7270b016f6b0efa55d7598fef6ef354ba77515956bb28e8e54/qtconsole-5.4.4-py3-none-any.whl (121 kB)
  121.9/121.9 kB 1.2 MB/s eta 0:00:00
Collecting jupyter-console (from jupyter)
```

3. Install JupyterLab

```
pip3 install -i https://pypi.tuna.tsinghua.edu.cn/simple jupyterlab
```

```
root@dofbot:/home/dofbot# pip3 install -i https://pypi.tuna.tsinghua.edu.cn/simple jupyterlab
Looking in indexes: https://pypi.tuna.tsinghua.edu.cn/simple
Collecting jupyterlab
  Downloading https://pypi.tuna.tsinghua.edu.cn/packages/31/7b/cd66f306c31a84a53c6a3a86e296586e8664f407a6ac5b7cfe6a433aa8c4/jupyterlab-2.2.9-py3-none-any.whl (7.9 MB)
  | 256 kB 499 kB/s eta 0:00:16
```

4. Switch back to the standard user account after installation is complete

```
su jetson
```

```
root@yahboom-virtual-machine:/home/yahboom# su yahboom
yahboom@yahboom-virtual-machine:~$
```

4. Configuring JupyterLab

1. Generate the configuration file

```
jupyter notebook --generate-config
```

```
yahboom@yahboom-virtual-machine:~/.jupyter$ jupyter notebook --generate-config
Overwrite /home/yahboom/.jupyter/jupyter_notebook_config.py with default config?
[y/N]y
Writing default config to: /home/yahboom/.jupyter/jupyter_notebook_config.py
yahboom@yahboom-virtual-machine:~/.jupyter$
```

2. Use IPython to generate the login password for JupyterLab.

ipython

At In [1], enter: `from notebook.auth import passwd`

At In [2], enter: `passwd()`

Then, enter the same password twice and press Enter to confirm. Note that this password is used for logging into the JupyterLab interface; for ease of recall, you may choose to keep it identical to your user account password. The system will then output the encrypted password hash; copy this entire block of text. Since the hash generated for each password may vary, please ensure you copy the specific text output by your system. In the In [3] cell, enter: `exit()`

```
yahboom@yahboom-virtual-machine:~/.jupyter$ ipython
Python 3.8.10 (default, May 26 2023, 14:05:08)
Type 'copyright', 'credits' or 'license' for more information
IPython 8.12.2 -- An enhanced Interactive Python. Type '?' for help.

In [1]: from notebook.auth import passwd

In [2]: passwd()
Enter password:
Verify password:
Out[2]: 'argon2:$argon2id$v=19$m=10240,t=10,p=8$13Ta0o+Xl7FAkUAS5FCqkg$yuuIX/MJ9
WIRw2g8wWXv/q4KsDMqFOWcURRi9mfz6M0'

In [3]: exit()
yahboom@yahboom-virtual-machine:~/.jupyter$
```

3. Configure the Jupyter configuration file

nano ~/.jupyter/jupyter_notebook_config.py

Scroll directly to the very end of the file and add the following content:

c.NotebookApp.ip = '0.0.0.0'

c.NotebookApp.open_browser = False

c.NotebookApp.password = [The entire hashed password string]

c.NotebookApp.port = 8888

```

this value
#           will correspond to the value of the Gateway url with 'ws' in place of
# 'http'. (JUPYTER_GATEWAY_WS_URL env var)
# Default: None
# c.GatewayClient.ws_url = None

#-----
# TerminalManager(LoggingConfigurable) configuration
#-----
##

## Timeout (in seconds) in which a terminal has been inactive and ready to be
culled.
#           Values of 0 or lower disable culling.
# Default: 0
# c.TerminalManager.cull_inactive_timeout = 0

## The interval (in seconds) on which to check for terminals exceeding the
# inactive timeout value.
# Default: 300
# c.TerminalManager.cull_interval = 300

c.NotebookApp.ip = '0.0.0.0'

c.NotebookApp.open_browser = False

c.NotebookApp.password =
'argon2:$argon2id$v=19$m=10240,t=10,p=8$13Ta0o+XL7FAkUAS5FCqkg$yuuIX/MJ9WIRw2g8wWXv/-
q4KsDMqFOWcURRi9mfz6M0'

c.NotebookApp.port = 8888

```

Finally, press Ctrl+S to save and Ctrl+X to exit.

5. Installing JupyterLab Extensions

1. Install Node.js and npm

```
sudo apt install nodejs npm
```

```

yahboom@yahboom-virtual-machine:~/jupyter$ sudo apt install nodejs npm
Reading package lists... Done
Building dependency tree
Reading state information... Done
nodejs is already the newest version (10.19.0~dfsg-3ubuntu1).
npm is already the newest version (6.14.4+ds-1ubuntu2).
0 upgraded, 0 newly installed, 0 to remove and 277 not upgraded.
yahboom@yahboom-virtual-machine:~/jupyter$

```

2. Install the Jupyter Widgets extension. Since this involves downloading and compiling, the process may take some time and could potentially encounter errors. If an error occurs, simply rerun the installation command.

```
sudo jupyter labextension install @jupyter-widgets/jupyterlab-manager
```

```

yahboom@yahboom-virtual-machine:~/jupyter$ sudo jupyter labextension install @jupyter-widgets/jupyterlab-manager
usage: jupyter [-h] [--version] [--config-dir] [--data-dir] [--runtime-dir]
               [--paths] [--json] [--debug]
               [subcommand]

Jupyter: Interactive Computing

positional arguments:
  subcommand      the subcommand to launch

optional arguments:
  -h, --help            show this help message and exit
  --version             show the versions of core jupyter packages and exit
  --config-dir          show Jupyter config dir
  --data-dir           show Jupyter data dir
  --runtime-dir         show Jupyter runtime dir
  --paths              show all paths
  --json              show JSON
  --debug             show debug output

```

3. Install the Status Bar extension

```
sudo jupyter labextension install @jupyterlab/statusbar
```

```

yahboom@yahboom-virtual-machine:~/jupyter$ sudo jupyter labextension install @jupyterlab/statusbar
usage: jupyter [-h] [--version] [--config-dir] [--data-dir] [--runtime-dir]
               [--paths] [--json] [--debug]
               [subcommand]

Jupyter: Interactive Computing

positional arguments:
  subcommand      the subcommand to launch

optional arguments:
  -h, --help            show this help message and exit
  --version             show the versions of core jupyter packages and exit
  --config-dir          show Jupyter config dir
  --data-dir           show Jupyter data dir
  --runtime-dir         show Jupyter runtime dir
  --paths              show all paths
  --json              show JSON
  --debug             show debug output

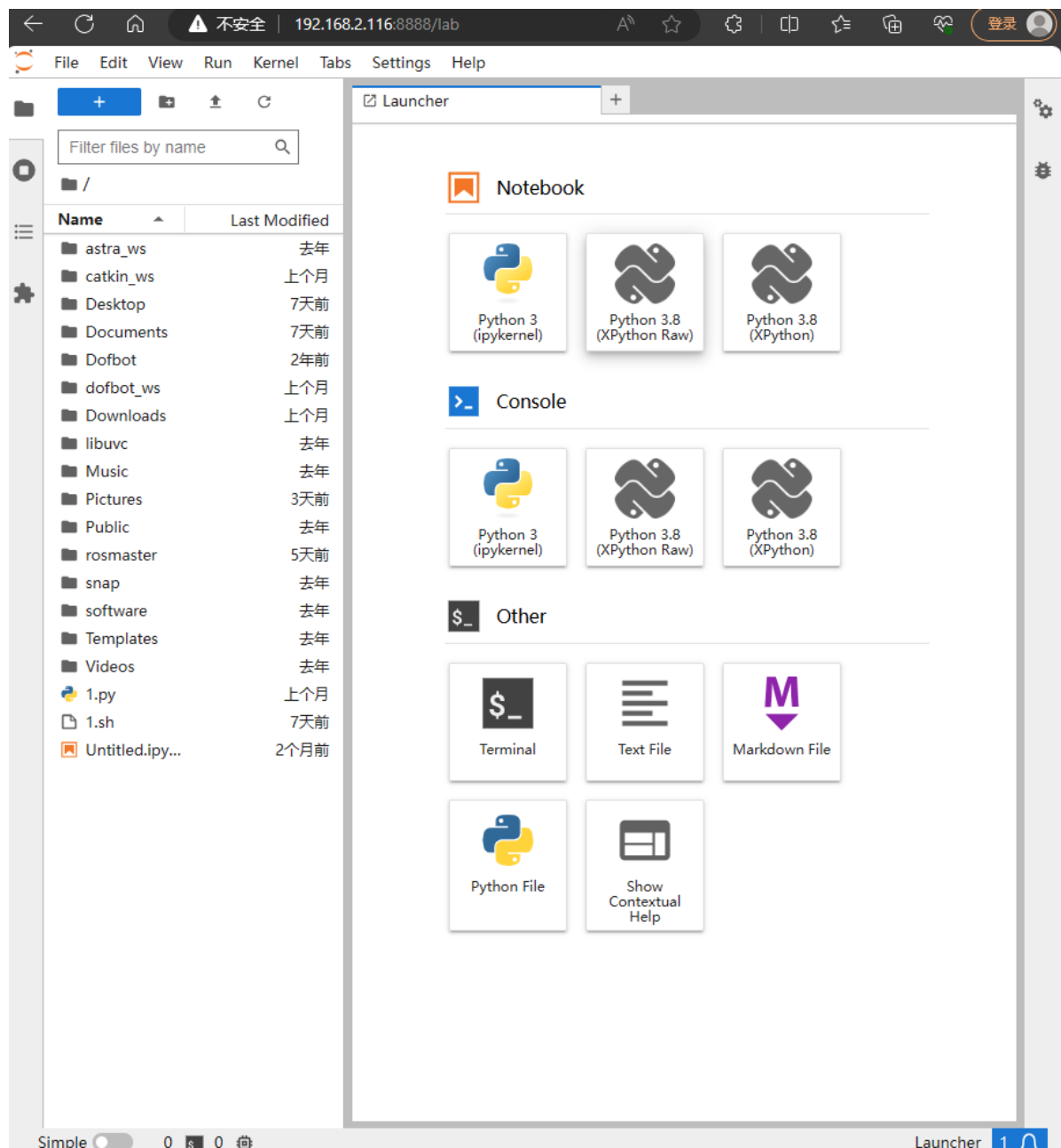
```

4. At this point, the installation of JupyterLab is complete.

6. Launching JupyterLab

1. Navigate to the directory where you intend to run your code. For this example, we will navigate to the root directory.
2. To launch JupyterLab, open a terminal window and enter the following command:

```
jupyter lab
```

8. Exiting JupyterLab

To exit JupyterLab, simply press **Ctrl+C** twice in the terminal window where you originally launched the application.